

Patch-Based Tokenisation of Discrete-Time Signals

Coursework — Points 1 & 2

Contents

1	Point 1	3
1.1	Setup and Formal Definition of the Tokenisation Operator	3
1.2	The Mechanics of Phase-Locking and Degeneracy	3
1.3	The Consequence: Loss of Observability	4
2	Point 2	5
2.1	The Patch-Induced Frequency	5
2.2	Phase-Locking Condition	5
2.3	Effect of Patch Size P	5
2.4	Token-Space Degeneracies	5
2.5	Overlap, Stride, and Blind-Spot Frequencies	6
2.6	Summary	6

1 Point 1

1.1 Setup and Formal Definition of the Tokenisation Operator

To rigorously understand the failure modes of patch-based architectures like Chronos-Bolt, we must first formally define the tokenisation process. Let $x[n]$ represent a discrete-time, univariate signal where the index $n \in \mathbb{Z}$ is sampled at a continuous frequency of f_s .

Unlike classical autoregressive models that process temporal data one scalar time-step at a time, modern foundation models employ patch-based tokenisation. This technique groups contiguous samples to radically reduce the effective sequence length and improve the computational efficiency of the transformer’s attention mechanisms. This is achieved by sliding a window of fixed size P (the patch length) across the signal, moving forward by a step size of S (the stride).

We can extract this sequence of tokens as follows:

$$\mathbf{p}_k = (x[kS], x[kS + 1], \dots, x[kS + P - 1]) \in \mathbb{R}^P, \quad k \in \mathbb{N} \quad (1)$$

Definition 1 (Patch Tokenisation Operator). *We can formalise this extraction as a linear map acting on the bounded signal space. The map $\mathcal{T} : \ell^\infty(\mathbb{Z}) \rightarrow (\mathbb{R}^P)^\mathbb{N}$, as defined above, is referred to as the patch tokenisation operator. When collecting an arbitrary number of K patches from a signal vector $\mathbf{x}$, the operation can be represented as a matrix multiplication:*

$$\mathbf{P} = \mathcal{T}\{\mathbf{x}\} = \mathbf{W} \cdot \mathbf{x} \quad (2)$$

Here, $\mathbf{W} \in \{0, 1\}^{KP \times N}$ is a block-binary selection matrix. Its k -th block consists of P rows that act as a mask, picking out the exactly P consecutive signal samples starting at the temporal index kS .

1.2 The Mechanics of Phase-Locking and Degeneracy

The fundamental vulnerability of the patch tokenisation operator emerges when the periodic structure of the input signal aligns perfectly with the model’s fixed stride grid. We refer to this specific synchronization as “phase-locking.”

Proposition 1 (Condition for Token Identity). *Two consecutive patches, $\mathbf{p}_k$ and $\mathbf{p}_{k+1}$, will be mathematically identical if and only if the signal amplitude remains constant across the stride shift for every element within the observation window:*

$$x[n] = x[n + S] \quad \forall n \in \{kS, kS + 1, \dots, kS + P - 1\} \quad (3)$$

Globally, if the underlying continuous signal is a pure sinusoid (or any periodic signal) with a fundamental period of T_0 samples, this exact alignment occurs whenever the stride S spans an integer multiple of the signal’s period:

$$\boxed{S = m \cdot T_0, \quad m \in \mathbb{Z}^+} \quad (4)$$

By substituting $T_0 = f_s/f$, we can express this condition in terms of the signal’s continuous frequency f and the sampling frequency f_s :

$$f = m \cdot \frac{f_s}{S}, \quad m \in \mathbb{Z}^+ \quad (5)$$

Proof. To satisfy $\mathbf{p}_k = \mathbf{p}_{k+1}$ strictly component-wise, it is required that $x[kS + j] = x[(k+1)S + j]$ for every internal patch index $j \in \{0, \dots, P-1\}$. For a periodic signal defined by $T_0 = f_s/f$, this strict equality holds true whenever the stride shift S represents a traversal of exactly m full wave cycles. In such instances, the sliding window captures the exact same geometrical segment of the wave at every step. $\square$

1.3 The Consequence: Loss of Observability

When the phase-locking condition $f = m \cdot (f_s/S)$ is met, we achieve a state where $\mathbf{p}_k = \mathbf{p}_{k+1}$ for all possible values of k . The token matrix $\mathbf{P}$ undergoes a mathematical rank collapse, reducing its rank to exactly 1.

This creates a “degenerate token sequence”, which induces severe representational failures within the network:

- **Complete Temporal Blindness:** Because the transformer processes a sequence of completely identical vectors, the temporal variation of the actual underlying signal becomes entirely invisible to the model’s self-attention mechanisms. The model cannot perceive that a wave is oscillating; it simply perceives a constant, flat embedding.
- **Observational Equivalence and Aliasing:** Any two distinct signals that differ only by a phase shift of S samples will produce the exact same sequence of tokens. They become observationally indistinguishable to the decoder, making it impossible for the model to correctly reconstruct or forecast the true phase of the original signal.
- **Soft Degradation at Boundary Frequencies:** This failure is not isolated to exact integer multiples. Even when patches are not perfectly identical but are merely highly correlated (i.e., when $S \approx mT_0$), the sequence suffers a *soft* loss of observability. The effective information bandwidth is severely restricted, explaining why predictive accuracy degrades smoothly around these critical frequency boundaries before fully collapsing.

For example, in the specific architecture of Chronos-Bolt, the sampling frequency is $f_s = 512$ Hz and the stride is $S = 16$. According to our derived condition $f = m \cdot (512/16)$, the fundamental patch frequency is precisely 32 Hz. Consequently, any input signal at 32 Hz, 64 Hz, 96 Hz, or other integer harmonics will trigger this precise structural aliasing, resulting in an immediate and sharp collapse in the model’s predictive accuracy.

2 Point 2

2.1 The Patch-Induced Frequency

The tokeniser introduces a characteristic *patch frequency*:

$$f_{\text{patch}} = \frac{f_s}{S}. \quad (6)$$

This is the rate at which new patches are produced (in Hz, referred to original time), analogous to a resampling rate in the token domain.

2.2 Phase-Locking Condition

Consider a sinusoidal signal $x[n] = A \cos(2\pi f n / f_s + \phi)$.

Definition 2 (Phase-Locking). *The signal phase-locks to the patch grid when its period $T_0 = f_s / f$ (in samples) is commensurate with S :*

$$\frac{S \cdot f}{f_s} \in \mathbb{Q}. \quad (7)$$

The degenerate (critical) case occurs when this ratio is an integer:

$$f = m \cdot \frac{f_s}{S} = m \cdot f_{\text{patch}}, \quad m \in \mathbb{Z}^+, \quad (8)$$

i.e. f is a **harmonic** of f_{patch} .

2.3 Effect of Patch Size P

A single patch of P samples spans a duration P / f_s seconds and covers $P \cdot f / f_s$ cycles of the signal. When

$$\frac{P \cdot f}{f_s} \in \mathbb{Z}^+, \quad \text{i.e.} \quad f = k \cdot \frac{f_s}{P}, \quad k \in \mathbb{Z}^+, \quad (9)$$

the patch contains an exact integer number of cycles: its content is *internally redundant*.

The most severe degeneracy occurs when both the stride and the patch-size conditions hold simultaneously:

$$f = \frac{\ell \cdot f_s}{\text{lcm}(P, S)}, \quad \ell \in \mathbb{Z}^+. \quad (10)$$

2.4 Token-Space Degeneracies

At $f = m f_s / S$ (harmonics of f_{patch}).

- All patches are translates of the same waveform by a phase that is a multiple of 2π : they become **identical** (cf. Point 1).
- The token sequence is constant; it encodes **no phase information**.
- Two signals at the same frequency but different phases (shifted by any multiple of S samples) map to the same token sequence: *token-space invariance*.

At $f = k f_s / P$ (but not harmonics of f_{patch}).

- Each patch is internally periodic, so its content is *within-patch redundant*.
- Inter-patch variation still exists (partial degeneracy).

2.5 Overlap, Stride, and Blind-Spot Frequencies

When $S < P$ (overlapping patches), consecutive patches share $P - S$ samples, partially mitigating the aliasing effect even near harmonic frequencies. For $S \geq P$ (no overlap or gaps), the aliasing is maximal; the **blind-spot frequencies** are:

$$f_{\text{blind}} = m \cdot \frac{f_s}{S}, \quad m = 1, 2, 3, \dots, \left\lfloor \frac{S}{2} \right\rfloor. \quad (11)$$

The upper limit follows from the Nyquist criterion applied to the token sequence, whose effective sample rate is f_s/S . This consequence followed what expressed in [?]

2.6 Summary

Condition	Effect
$f = m f_s / S$	Full phase-locking $\rightarrow$ identical patches $\rightarrow$ constant token sequence
$f = k f_s / P$	Internal patch periodicity $\rightarrow$ within-patch redundancy
Both simultaneously	Maximal degeneracy
$S < P$ (overlap)	Partial mitigation; residual phase information
$S \geq P$ (no overlap)	Full degeneracy at blind-spot frequencies

Table 1: Summary of conditions and their effects on token-space observability.

References